Tech Science Press

http://doi.org/10.32604/jimh.2026.000000

ARTICLE

Lesion-grounded prototype meta-learning for few-shot thyroid ultrasound: segmentation, diagnosis, and faithful explanations under domain shift

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Version September 6, 2026 submitted to JIMH

ABSTRACT: Most studies report thyroid-ultrasound computer-aided diagnosis as single-benchmark classification, leaving annotation cost, cross-domain transfer, and explanation reliability unmeasured. We present LG-PM (lesion-grounded prototype meta-learning), a pipeline that couples few-shot support-conditioned nodule segmentation with malignancy classification from frozen lesion descriptors. We evaluate it in a pre-registered, audit-trailed programme across five public cohorts (TN3K, TNUI-2021, DDTI, TN5000, ThyUS2Path). Zero-shot segmentation transfers to unseen domains without any target annotations, coming within 3.9–6.1 Dice points of a fully supervised external reference. Segmentation performance saturates at five support masks, reaching 98.6–99.5% of the twenty-mask score. In-domain, lesion grounding improves on an identical whole-image head (+0.0368 AUROC), though not above the fine-tuned ceiling. Cohort ranking collapses under external shift for every model (AUROC 0.478–0.548 on 839 pathology-confirmed ThyUS2Path cases). The frozen operating point nonetheless transfers: the grounded head retains 15.9% sensitivity at 90% specificity where the fine-tuned classifier retains 1.3% (McNemar 77 vs 1, $p<0.0004$). Explanations exhibit the same pattern. The grounded exact attribution localises an order of magnitude more precisely than class-activation-map (CAM) comparators (ratio 41.6 vs 3.6, $p<0.0004$). Deletion-based faithfulness metrics instead favour diffuse-evidence models — a property of the metric family that we characterise rather than a finding we omit. Selective prediction referring 50% of cases removes ≈86% of malignant misses at the frozen point. Lesion grounding, not cohort ranking, is what survives domain shift.

KEYWORDS: thyroid ultrasound; few-shot segmentation; prototype learning; domain shift; external validation; explainability; selective prediction

1 Introduction

Thyroid-ultrasound computer-aided diagnosis is usually framed as a classification problem: given an image, decide whether the nodule is benign or malignant. In practice the diagnostic pipeline couples two visual tasks — locating the nodule against gland tissue, speckle and artefacts, and then assessing its morphology for malignancy risk. A classifier that is accurate only on perfectly cropped lesions does not represent a complete workflow; a segmenter that localises without supporting risk assessment is equally incomplete. We therefore frame the problem jointly and annotation-efficiently: *few-shot nodule segmentation* supplies support–query localisation from a handful of annotated cases, and the resulting lesion representation feeds benign/malignant classification.

Four unresolved challenges motivate the study: (i) public thyroid-ultrasound datasets are heterogeneous in label source, annotation type, patient structure, scanner distribution and clinical endpoint; (ii)

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segmentation and classification are studied separately even though segmentation errors propagate directly into classification; (iii) few-shot methods developed on natural images cannot be assumed to work on low-contrast lesions with speckle, small foregrounds and high intra-class variability; and (iv) explanation heatmaps are routinely displayed without testing lesion overlap, stability under domain shift, or faithfulness to the decision [1].

The dataset ecosystem maps onto these challenges. DDTI provides early open-access expert-annotated nodules [2]; TN3K a larger multi-device open segmentation set [3]; TN-SCUI2020 class labels with delineations [4]; TN5000 5,000 biopsy-confirmed B-mode images with expert annotations and an official 7:1:2 split [5]; TNUI-2021 a three-cohort external segmentation battery; and ThyUS2Path pathology-confirmed case-level labels for 8,508 images from 842 cases [6]. These resources are complementary, not interchangeable: masks, boxes, TI-RADS-style attributes [7] and pathology labels answer different clinical questions, and a realistic pipeline must use each annotation where it exists — a *partially supervised* design in which segmentation loss applies only to mask-labelled data and classification loss only to pathology-labelled data, sharing one representation.

The literature supports five conclusions: the task is inherently one of joint localisation and malignancy assessment, and recent multicentre thyroid-ultrasound AI studies increasingly evaluate the two together [8–11]; dataset roles must be documented rather than merged; few-shot segmentation offers a principled lens on dense-label efficiency but must be adapted to ultrasound; predicted masks can propagate errors, requiring explicit ground-truth versus predicted comparisons; internal–external performance declines are now quantified across radiology studies of 2022–2025 [12]; and explanations should be evaluated inside the segmentation–classification pipeline on overlap, faithfulness, stability and usefulness. On that basis nine gaps remain open (i.e. not answered by any reviewed study), consolidated here: **Gap 1**, few-shot thyroid-nodule segmentation is not characterised across matched budgeted episodes; **Gap 2**, segmentation and classification are rarely evaluated as one support-conditioned pipeline; **Gap 3**, error propagation from predicted masks is underreported; **Gap 4**, heterogeneous supervision (masks, boxes, pathology) is not fully exploited without conflating label meanings; **Gap 5**, few-shot segmentation under ultrasound domain shift is underexplored; **Gap 6**, support/mask uncertainty is not connected to clinical referral; **Gap 7**, explanation evaluation is not integrated with segmentation; **Gap 8**, reproducible external benchmarks (episode manifests, patient-level splits, frozen thresholds) are rare — a deficit the community has now formalised in reporting guidance (TRIPOD+AI [13], STARD-AI [14], CLAIM [15]) and in trust/deployment consensus (FUTURE-AI [16]); **Gap 9**, edge feasibility is rarely co-evaluated with annotation-efficient medical AI.

Meta-learning supplies the machinery: prototypical networks [17], siamese and matching networks [18, 19], MAML-style optimisation [20], and few-shot segmentation decoders (PANet, PFENet, HSNet) [21–23], with distributional-representation heads such as REPRI [24] nearest in spirit. Universal in-context segmenters (UniverSeg [25]) and promptable medical foundation models (MedSAM [26]) have since shown that task-free adaptation is feasible in principle, although structured evaluations caution that zero-shot medical transfer remains uneven [27]. Our design differs in that the support-conditioned mask is not an add-on but the *bridge* between heterogeneous datasets and pathology-level classification — evaluated under shift with explanation reliability measured, not illustrated [28,29].

The overall aim is a cross-domain, few-shot, lesion-grounded framework that segments thyroid nodules from limited support annotation and uses the lesion representation for benign/malignant classification.

1.1 Research questions and hypotheses

The central research question — *can few-shot, support-conditioned nodule segmentation improve cross-domain thyroid-ultrasound malignancy classification while providing calibrated, anatomically grounded, and faithful explanations?* — decomposes into nine research questions with nine pre-registered hypotheses, consolidated verbatim in Table 1.

Table 1: Research questions and pre-registered hypotheses (locked wording, abridged to the test clause; full statements in the pre-registered analysis plan). RQ9/H9 (edge feasibility) was descoped after Phase 6 (Methods, 'Scope') and is retained as future work.

	Research question	Hypothesis (test clause)
RQ1	Few-shot segmentation accuracy from K annotated supports, $K \in \{1, 5, 10, 20\}$	H1: support conditioning achieves useful localisation with fewer labels than per-domain training/fine-tuning
RQ2	Transfer of few-shot segmentation to unseen domains vs supervised baselines	H2: domain-diverse episodic training improves external Dice/IoU/boundary accuracy over single-domain baselines
RQ3	Does a predicted mask improve classification over whole-image/ROI-independent prototypes?	H3: predicted-mask lesion features improve external AUROC/AUPRC/sensitivity or calibration
RQ4	Performance lost when ground-truth masks are replaced by predicted masks/boxes	H4: the gap shrinks with support size, and uncertainty identifies the largest-gap cases
RQ5	Can mask-labelled and pathology-labelled datasets be optimised jointly without shortcuts?	H5: joint training improves cross-domain performance over either annotation type alone
RQ6	Does peri-lesional context add discrimination, calibration, robustness?	H6: lesion+context beats lesion-only, with less attribution outside the lesion
RQ7	Can uncertainty drive referral and cut malignant false negatives at useful coverage?	H7: abstaining on high-uncertainty cases reduces malignant false negatives at useful coverage
RQ8	Are pipeline explanations more lesion-aligned, stable, faithful than unconstrained CAMs?	H8: grounded explanations show higher lesion overlap and stability without lower faithfulness
RQ9	Trade-off of performance vs size/memory/latency/energy at the edge	H9: a distilled, quantised student preserves a preset performance fraction (descoped)

The scope is deliberately bounded: the primary task is binary benign/malignant classification of thyroid nodules in B-mode ultrasound, supported by few-shot segmentation. We do not claim that segmentation or classification replaces pathology, radiologist assessment, or TI-RADS-based management; we evaluate whether an annotation-efficient localisation module improves a well-defined malignancy-classification problem under dataset and scanner shift. Four methodological boundaries are held throughout: support/query disjointness at patient or nodule level; no external data used for tuning, threshold selection, calibration, or explainer selection; segmentation and classification metrics reported separately before integration; and ground-truth masks always distinguished from predicted masks.

2 Material and Methods

2.1 Study design and methodological objective

This study investigates whether few-shot nodule segmentation can provide an annotation-efficient, anatomically grounded pathway to thyroid-nodule malignancy classification under ultrasound domain shift. The proposed *Lesion-Grounded Prototype Meta-learning* (LG-PM) framework connects four components:

(i) support-conditioned nodule segmentation; (ii) lesion–context feature extraction; (iii) lesion-grounded classification from frozen descriptors; and (iv) explanation analysis. A supervised segmentation reference is trained first (Stage 1), the support-conditioned module built on its frozen encoder (Stage 2), and classification built on the frozen segmentation machinery (Stage 3): grounding effects are measured against identical features, splits, and evaluation code, and no classification-domain gradient enters segmentation.

The two endpoints are reported separately and never proxy for each other: query-image nodule localisation under a limited number of labelled support masks (segmentation), and benign-versus-malignant discrimination on pathology-labelled cases (classification). Evaluation runs under five regimes — in-domain, zero-shot cross-domain, and few-shot target-adapted segmentation; in-domain and strictly zero-shot external classification — with per-episode support/query disjointness, source-validation-only selection and calibration, and pooling permitted only where pre-declared (Section 2.10). A full-capacity teacher (ResNet34-encoder U-Net) serves the main scientific evaluation.

2.2 Data sources and annotation roles

Datasets are used according to their available annotations, not merged into one pool. TN3K (3,493 image–mask pairs) [3,30] is the source segmentation domain; TNUI-2021 (1,381 pairs) [31] the primary external segmentation domain, its training split additionally donating support cases; DDTI (637 pairs, accessed through a version-documented mirror) [2,32] a secondary external segmentation domain. TN5000 (5,000 pathology-labelled images with expert bounding boxes) [5,33] is the source classification domain, official split retained. External classification evaluation comprises the three official TNUI-2021 cohorts (test, validation, training; pure query pools), the XML-labelled DDTI subset, and — at case level — the pathology-labelled ThyUS2Path cohort [6,34]; TN-SCUI2020 remains an optional extension pending authorised access. Every image is recorded with annotation provenance, split assignment, and a SHA-256 duplicate-check hash (field list and annotation hierarchy: Supplementary Sections S1.1–S1.2). TN3K, TNUI-2021, and DDTI provide no usable patient linkage (a recorded limitation), so their image-level results are interpreted as potentially correlated; no target-domain test annotation informs any design, tuning, or calibration decision. All cohorts are fully de-identified public datasets released for research use; institutional ethics-committee approval and participant consent were not required for this secondary analysis.

Classification labels enter through three documented parsers (full precedence specification: Supplementary Section S1.3). TN5000: *malignant-if-any* across annotated objects, the oracle box their union. TNUI-2021: the polarity (1 = malignant) is undocumented and adopted as an *assumption-with-evidence* — the alternative would force in-domain-validated features to anti-classify on every TNUI cohort, and cohort sizes and prevalence are consistent with the published dataset description — recorded in the limitations rather than silently absorbed. DDTI: per-case XML diagnoses follow a fixed precedence (literal strings over numeric codes over TI-RADS), aggregated malignant-if-any, with a per-run vote profile for audit.

2.3 Data splitting, de-duplication, and external-validation protocol

Every image is SHA-256 hashed; exact duplicates are removed once per dataset (highest-priority partition retained), and hashing also covers every support-donor pool and every classification-cohort pair (mechanics: Supplementary Section S1.4). Each notebook re-derives the clean manifests from raw files and verifies them at hash level against the locked audit manifests; classification phases must additionally re-derive the locked TN5000 split sizes before any model runs. Official partitions are retained: TN3K trainval is split 85/15 by image identifier (fixed seed 42); the TN5000 validation split alone serves early stopping,

operating-point selection, and calibration; all three TNUI-2021 partitions are external query pools; DDTI is used whole. Support/query assignments are disjoint per episode, enforced by hash at sampling time.

Segmentation transfer is reported as *zero-shot* (no labelled target cases; supports from the source training pool) or *mask-shot* (K_m labelled target pairs as supports: TNUI-2021 donors from its train split, DDTI donors from the query pool with per-episode disjointness). Budget comparisons use nested support sets:

$$\mathcal{S}_1 \subset \mathcal{S}_5 \subset \mathcal{S}_{10} \subset \mathcal{S}_{20}, \tag{1}$$

so budget effects are measured on the same query cases, paired by construction. Classification transfer is strictly zero-shot — no external annotation, not even unlabelled adaptation, alters any learned object — and all split and episode manifests are saved for every run.

2.4 Episode formulation

A segmentation episode pairs a support set of K_m image–mask pairs, $K_m \in \{1, 5, 10, 20\}$ nested as in Eq. (1), with $Q_m = 8$ disjoint query images (formal definitions: Supplementary Section S3.1); multiple nodules per image form a single foreground union. Meta-training samples 300 episodes per epoch from the source pool, re-drawing K_m uniformly from $\{1, 5, 10, 20\}$ at every episode; validation uses 100 episodes and evaluation 200 per condition.

The fixed support episode.

All classification cohorts (TN5000 train/val/test and every external cohort) are segmented *zero-shot* with one fixed support episode: a single deterministic draw of $K_m = 20$ image–mask pairs from the TN3K internal training pool under a fixed seed, archived by case identifier in the first classification phase. Every later phase re-derives the episode and *asserts identifier-level equality* before any descriptor is computed. Predicted-mask descriptors therefore require no segmentation annotation in any classification domain, and the segmentation machinery — weights, supports, preprocessing — is identical across all cohorts and phases.

Classification label budgets.

Classification supervision is a *label-budget* analysis on the source domain only: primary classifiers (Section 2.5.4) train on the full source training split, and a nested per-class sweep $K_c \in \{5, 10, 20, 50, 200, \text{all}\}$ (25 deterministic resamples per finite budget) instantiates class-prototype heads from the same descriptors — the classification-side annotation-efficiency curve.

Partially supervised objective (pre-registered joint phase).

The joint fine-tuning phase (Section 2.7.1) mixes batch types and activates only the losses each sample supports (I_m: mask available; I_y: malignancy label available):

$$\mathcal{L}_{\text{partial}} = I_m(x)\lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + I_y(x)\lambda_{\text{cls}}\mathcal{L}_{\text{cls}}. \tag{2}$$

2.5 LG-PM architecture

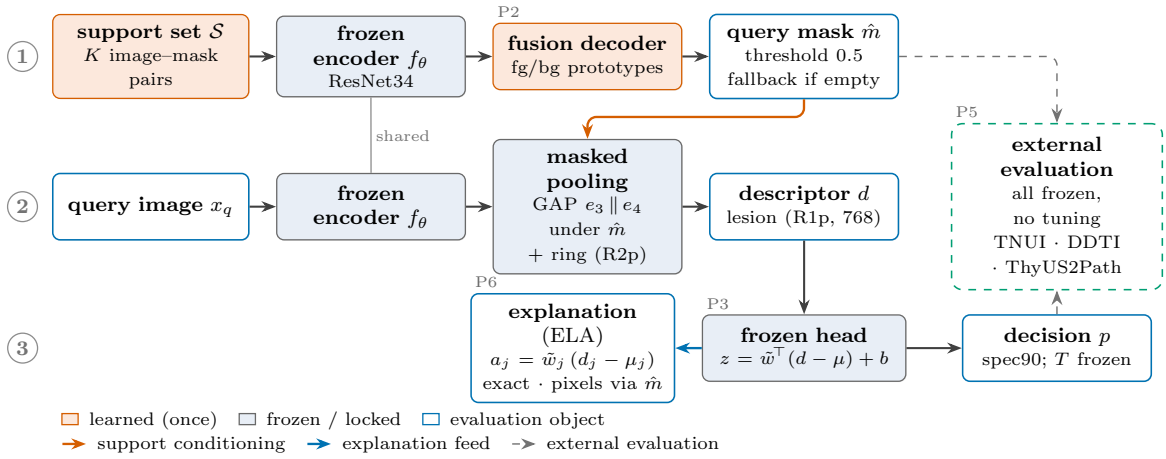


Figure 1: The LG-PM pipeline. Top: a frozen support episode of K nested image–mask pairs from TN3K-train conditions the prototype fusion decoder (trained once, Phase 2), which predicts the query mask $\hat{m}$ at threshold 0.5 with a whole-image fallback when the mask is empty (rates in Supplementary Table S17). Middle: the query image passes the same frozen ResNet34 encoder (shared weights, vertical link), and masked pooling averages its $e_3 \,|\, e_4$ features under $\hat{m}$ (plus a 15-px context ring in the R2p ablation), giving the lesion descriptor d (R1p, 768-d; image-level R0 and lesion+ring R2p 1536-d variants: Supplementary Table S4). Bottom: one linear head trained once on TN5000-train, $z = \tilde{w}^\top(d-\mu)+b$, gives $p = \sigma(z/T)$ with T fitted on validation only and the operating point frozen at 90% specificity (raw–raw convention); the exact linear attribution $a_j = \tilde{w}_j(d_j - \mu_j)$, $\sum_j a_j + c = z$, is projected to pixels through $\hat{m}$ (Phase 6). Circled numerals mark the three stages (1 support conditioning, 2 query encoding, 3 frozen readout and attribution); the vermillion arrow is the support-conditioning link, the blue arrow the explanation feed, and dashed arrows enter the external-evaluation frame. P-badges mark the phase that produced each object. Dashed frame: external evaluation on TNUI-2021, DDTI and ThyUS2Path with every object frozen — no target tuning, no target calibration.

The architecture couples a shared encoder, a support-conditioned segmentation module, a lesion–context descriptor module, and a family of classification heads (Fig. 1); the teacher's encoder is an ImageNet-pretrained ResNet34 inside a U-Net-style decoder. The working resolution is 384 × 384; the prototype module uses levels $\{e_2, e_3, e_4\}$ (strides 8/16/32; 128/256/512 channels) and descriptors pool $\{e_3, e_4\}$ (formal definitions: Supplementary Section S3.1). Images are letterboxed; masks share the geometric transform (nearest-neighbour) and are binarised at grey level ≥ 127, selected by visual audit against > 0 and ≥ 250 alternatives; inputs are ImageNet-normalised.

2.5.1 Stage 1 optimisation (supervised reference)

The Stage-1 reference is a ResNet34-encoder U-Net trained with the Tversky-based objective below (optimiser, scheduler, augmentation, and threshold-sweep listing: Supplementary Section S2.1). Each epoch selects the better of raw and exponential-moving-average (EMA) weights by validation Dice. The sigmoid operating threshold is tuned on the internal validation split only and frozen before any test or external evaluation; test-time augmentation (TTA) is reported alongside single-view inference. The tuned threshold value is reported in Results.

2.5.2 Support-conditioned segmentation prototypes

Each support mask is downsampled (nearest-neighbour) to the spatial size of feature level l; the foreground prototype is the masked average

$$p_f^l = \frac{1}{Z_f^l} \sum_{i=1}^{K_m} \sum_u \bar{m}_i^l(u) F^l(x_i, u), \qquad Z_f^l = \sum_{i=1}^{K_m} \sum_u \bar{m}_i^l(u) + \epsilon, \tag{3}$$

and the background prototype uses the complementary mask:

$$p_b^l = \frac{1}{Z_b^l} \sum_{i=1}^{K_m} \sum_u (1 - \bar{m}_i^l(u)) F^l(x_i, u). \tag{4}$$

Query pixel features are scored by the negative squared Euclidean distance,

$$s_c^l(x_q, u) = -\left\| F^l(x_q, u) - p_c^l \right\|_2^2, \qquad c \in \{f, b\}, \tag{5}$$

score maps are bilinearly resized to the stride-8 grid (48×48) and stacked as $2L$ channels, and a lightweight convolutional decoder D_ϕ ($2L \to 32 \to 16 \to 1$) produces the query mask:

$$\hat{m}_q = \sigma\big(\mathrm{up}\big(D_\phi\big(\{s_f^l, s_b^l\}_l\big)\big)\big). \tag{6}$$

The *direct* ablation replaces D_ϕ with the parameter-free sum $\sum_l (s_f^l - s_b^l)$, isolating the learned fusion; Stage 2 freezes the encoder (BatchNorm in inference mode) and trains only D_ϕ. A variance-weighted uncertainty extension is pre-specified (equations: Supplementary Section S3.2) but was not executed: the point-prototype model is the primary and only reported configuration.

2.5.3 Lesion–context descriptor regimes

The representation module pools encoder features over five pre-specified regions: for any binary mask $\Omega \in \{0, 1\}^{H \times W}$ the descriptor is the masked global average over levels $l \in \{e_3, e_4\}$,

$$z_\Omega(x) = \Big\|_{l \in \{e_3, e_4\}} \frac{\sum_u \tilde{\Omega}^l(u)\, F^l(x, u)}{\max\big(\sum_u \tilde{\Omega}^l(u),\, 1\big)}, \tag{7}$$

computed per image (never across the batch). Table 2 defines the five regimes and the governing conventions, all fixed before any evaluation.

2.5.4 Classification heads

Three head families act on the cached descriptors; all supervision is source-domain (TN5000 train) only.

Table 2: Lesion–context descriptor regimes (the five pre-specified pooling regions). Each is a masked global average over encoder levels $\{e_3, e_4\}$ (Eq. (7)), computed per image — never across the batch — and cached once per cohort in the feature bank.

Regime	Pooled region Ω (and fallbacks)	Role
R0	whole image (plain global average)	ungrounded reference
R1p	$\mathbf{1}[\hat{m} \geq 0.5]$, the binarised prediction from the fixed support episode	deployable lesion grounding
R2p	lesion plus peri-lesional ring (elliptical dilation, radius 15 px at 384^2, minus the mask; empty ring falls back to the image complement)	grounding + context
R1g	oracle region (TN5000 expert box; TNUI-2021/DDTI ground-truth mask)	oracle lesion; error-propagation pricing
R2g	oracle lesion plus its ring	grounding + context, no segmentation error

Governing conventions. (1) *Empty-mask fallback*: fewer than 4 predicted pixels, and R1p/R2p pool the whole image (identical to R0 for that case); the case is flagged and the rate reported per cohort. (2) *Oracle discipline*: oracle regions appear only in the labelled oracle rows and the error-propagation analysis, never in a deployable path. (3) *Feature bank*: all five regimes are computed once per cohort and cached with per-case quality fields (Supplementary Section S2.6); the cache is the unit of reuse and tolerance verification. (4) *No learned fusion*: the ring contrast (R1 versus R2) is reported explicitly as the context-utility ablation.

Linear heads (primary).

Each regime's descriptor is standardised with training-split statistics and fitted with a logistic linear head under weighted binary cross-entropy (training and selection protocol: Supplementary Section S2.3). The only quantity that varies across regimes is *what is pooled*, so R0–R1p contrasts isolate grounding and R1p–R1g contrasts isolate segmentation error.

Class-prototype heads (annotation-efficiency analysis).

Descriptors are L2-normalised, and per-class means over nested per-class draws from the training split ($K_c \in \{5, 10, 20, 50, 200, \text{all}\}$, 25 resamples per finite budget, fixed seed) are scored by the cosine margin

$$s(x) = \cos\big(z(x), \mu_1\big) - \cos\big(z(x), \mu_0\big), \tag{8}$$

reported as mean ± standard deviation of test AUROC across draws; in-domain only.

Whole-image ERM baseline (the fair comparator).

The comparator is whole-image empirical risk minimisation (ERM): an ImageNet-pretrained ResNet34 fine-tuned end-to-end on TN5000 (regimen: Supplementary Section S2.4). This row must be strong — it is the conventional alternative a reviewer compares the grounded pipeline against. A shrunk-covariance uncertainty scorer is a designated, unreported extension.

2.6 Training objectives

The Stage-1 reference uses binary cross-entropy (BCE) plus a soft Tversky term,

$$\mathcal{L}_{\text{seg}}^{(1)} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Tversky}}, \qquad \mathcal{L}_{\text{Tversky}} = 1 - \frac{\text{TP} + \epsilon}{\text{TP} + \alpha\,\text{FP} + \beta\,\text{FN} + \epsilon}, \tag{9}$$

with $\alpha = 0.3$, $\beta = 0.7$, penalising false negatives on small, fuzzy-boundary nodules. The Stage-2 decoder trains episodically with query-pixel BCE plus a soft Dice term; the direct variant needs no fitting. Classification heads minimise BCE with positive-class weight $\omega = n_{\text{neg}}/n_{\text{pos}}$ from the source training split. A boundary loss and an augmentation-consistency term were considered and deferred.

2.7 Training procedure

Stage 1. The ResNet34-UNet is trained on all TN3K internal-training masks; the better of raw/EMA weights by validation Dice is checkpointed; no external-domain image participates. **Stage 2.** With the encoder frozen, the decoder alone is trained on source episodes (nested supports, eight queries, light joint augmentation applied consistently across images and masks), with early stopping on mean query Dice over the validation episodes (regimen: Supplementary Section S2.2); multi-source episodic training is a designated extension. **Stage 3.** The fixed support episode segments every TN5000 image zero-shot; the five regimes are pooled and cached; the heads, budget sweep, and ERM baseline are trained. This stage's checkpoint, cached probabilities, and head parameters are the sole classification artefacts permitted later. **Stage 4.** On source validation only: a scalar temperature per model (grid search; Supplementary Section S2.5), the decision threshold (maximal sensitivity at specificity ≥ 0.90 on raw validation probabilities), the segmentation threshold, and the TTA configuration — all frozen before any test or external evaluation. **Stage 5.** Frozen objects only cross domains, under Section 2.10.

2.7.1 Pre-registered joint partial-supervision phase (Phase 4)

Pre-registered before execution (full regimen: Supplementary Section S4.1). A multi-task trunk (Stage-1 checkpoint, *unfrozen*) is fine-tuned on interleaved TN3K segmentation (BCE + Tversky) and TN5000 classification (weighted BCE) batches: **JOINT-g** classifies from the trunk's *own* predicted masks; **JOINT-img** pools the whole image on the same trunk (grounding versus multi-tasking). The head is first warmed on the frozen trunk, then jointly fine-tuned at 10^{-5}, an order of magnitude below the ERM regimen; checkpoint selection maximises source validation AUROC *subject to a segmentation guard*, TN3K-validation Dice ≥ 0.8066 (the pre-registered non-collapse constant plus a 0.01 margin), and the test split never participates.

The guard exists because protocol v1 failed it: under a single-stage joint fine-tune with AUROC-only selection, the freshly initialised classification head's gradient rotated the shared encoder from the first epoch — saved-model TN3K-test Dice was 0.6686 against a gate of 0.7966 — while a pristine-checkpoint control reproduced the locked reference exactly, validating the harness. Should no joint epoch satisfy the guard, the best-Dice state is retained for diagnosis and the gate is allowed to fail: the joint phase is then reported as a negative result rather than analysed. Both arms share identical batch/augmentation streams (`set_seed(42)` first) and differ only in pooling region. The pre-declared in-domain family G is headed by G1, JOINT-g − ERM AUROC (is the fine-tuning gap closed?); G2–G4 (frozen versus joint grounding; grounding versus multi-tasking; operating-point contrast) are defined verbatim in Supplementary Section S4.1. The joint model then enters the frozen external battery, its thresholds and temperature fitted on its own TN5000 validation probabilities only, under the manifest, non-collapse, and ERM-sanity acceptance gates. A pre-registered uncertainty component reports coverage-versus-malignant-FNR selective-prediction curves from calibrated probabilities.

2.7.2 Case-level external validation on ThyUS2Path (Phase 5)

Pre-registered before execution. ThyUS2Path contributes histopathology-labelled cases from an acquisition network disjoint from all source pools (two release batches, multiple images per case); the question is whether the frozen pipeline and its pre-frozen operating point survive a severe cohort-level shift under a pathology gold standard. No metric cell runs before the audit gate, and no inferential statement is unsealed before the licence (full A1–A5 mechanics: Supplementary Section S4.2): per-batch label coverage must reach 95%; cases are scored by the unweighted mean of per-image calibrated probabilities (first-image view as descriptive sensitivity analysis); SHA-256 plus perceptual-hash screening against every support or training pool excludes a *whole* case on a single match; cohort gates require 300 assembled cases and prevalence within $[0.20, 0.85]$; and the operator attests label polarity against corpus documentation. The family is computed under the seal but reported WITHHELD (descriptive statistics only) unless the gate passes computationally *and* attestation is complete; the amendment fixing the two-table join, the case-mean primary unit, and whole-case exclusion was adopted 2026-08-22, strictly before any target-side forward pass. Pre-registered family J (case level; 5,000 paired bootstrap resamples; Bonferroni $\alpha = 0.05/4 = 0.0125$; raw-raw convention; nothing refitted) is headed by **J1**, sensitivity@spec90 R1p − ERM with McNemar discordants on malignant cases — the primary external operating-point replication; J2–J4 are defined verbatim in Supplementary Section S4.2. Oracle regimes are excluded by pre-declaration (no lesion annotation).

2.7.3 Pre-registered lesion-grounded explainability battery (Phase 6)

Pre-registered before execution; nothing is refitted and the only backward passes compute activation gradients. For every linear head (R0, R1p, R2p; JOINT-g, JOINT-img) the logit is exactly linear in the pooled features, so per-pixel attribution is computed in closed form within exactly the pooled regions (Exact Linear Attribution, ELA), asserted to reproduce each head's logit to $< 10^{-3}$; ERM uses Grad-CAM at layer4 with Eigen-CAM and Score-CAM robustness rows, all CAMs ReLU-ed and ℓ_1-normalised. Four axes (full specifications and perturbation recipes: Supplementary Section S4.3): *plausibility* — explanation concentration in the TN5000 expert box (localisation ratio: in-box explanation share over box-area share), plus descriptive pointing games; *faithfulness* — deletion (primary) and insertion curves against a per-image blur baseline and a random-ordering control, the gap their trapezoidal area difference; *stability* — Spearman correlation of clean and perturbed CAMs, full pipeline recomputed; *randomisation sanity* — cascading reinitialisation, self-gated at mean $|\rho| < 0.2$ at the final stage (asserted). Cohorts: TN5000-test (n=950, 693 malignant; expert boxes) and a seeded ThyUS2Path first-image panel (n=600, from the licensed Phase-5 839-case manifest; label-free metrics, independent of the Phase-5 licence). Pre-registered family K (image level; 5,000 paired bootstrap resamples; Bonferroni $\alpha = 0.05/4 = 0.0125$; seed 42) is headed by **K1**, the GT-localisation ratio R1p − ERM on TN5000-test (primary plausibility); K2–K4 are defined verbatim in Supplementary Section S4.3.

2.8 Baselines and ablation studies

All baselines share the same cleaning, splits, preprocessing, episode manifests, and seeds. Segmentation compares the full-data Stage-1 reference (TTA and single-view), parameter-free direct matching, the proposed fusion-decoder model, and literature-style few-shot architectures (PANet-, PFENet-, HSNet-like) under identical episodes, subject to feasibility. The six pre-declared classification rows span whole-image ERM, the R0/R1p/R2p grounding family, and the R1g/R2g oracle rows, supplemented by the prototype budget sweep (annotation efficiency) and the ring contrast (context utility). Further ablations: ROI source

(R0/R1g/R1p); ring on/off; nested budgets $K_m \in \{1, 5, 10, 20\}$ and K_c; zero-shot versus mask-shot; decoder versus direct; frozen versus joint trunk; the deferred consistency loss; the designated multi-source extension.

2.9 Evaluation metrics

Dense segmentation evaluation reports per-case Dice, intersection-over-union (IoU), sensitivity, precision, and the 95th-percentile Hausdorff distance (HD95; non-empty masks only; definitions: Supplementary Section S3.3); episodic evaluation reports per-episode Dice and IoU. Aggregate 95% intervals use 2,000 bootstrap replicates (image level for dense, episode means for episodic, preserving episode pairing), and every external phase re-reports segmentation quality control against the locked Phase-2 zero-shot references before any classification table.

2.9.1 Classification metrics and decision conventions

Reported metrics: AUROC; AUPRC; sensitivity at the validation-frozen specificity-0.90 operating point; Brier score and expected calibration error (ECE, 15 equal-width bins) around temperature scaling; reliability curves for the headline rows. Four conventions were fixed after the paired-analysis audit: (1) thresholds and temperatures come from the source validation split only, frozen before any test or external evaluation; (2) the raw-raw convention never mixes raw-threshold binary decisions with calibrated probabilities (rank metrics are convention-invariant); (3) confidence intervals are case-level bootstrap, 2,000 resamples, percentile 2.5/97.5; (4) externally, calibration metrics are descriptive only — any deployment interpretation requires local recalibration, stated rather than performed.

2.9.2 Selective prediction

Referral scoring combines classification entropy and the probability margin from calibrated probabilities (a Phase-4 pre-registered component); risk–coverage curves report false-negative rate, sensitivity, and error at prespecified coverage — an assistive mechanism, not an autonomous rule.

2.10 External classification transfer protocol

This pre-registered protocol binds every external classification phase and prevents silent refitting: the frozen objects — encoder, fusion decoder, fixed support episode (identifier-asserted), descriptor code, head weights, thresholds, temperatures — are byte-identical to the source phase, and a battery of tolerance, sanity, quality-control, and replication gates is checked before any external table, so that every recomputed row must match its locked value within pre-set tolerances and pooling-dependent analyses are *withheld* rather than silently recomputed (full gate specifications, tolerances, and pooling policy: Supplementary Section S4.4). Exactly one pool is pre-declared (TNUI-2021 test + validation); later cohorts are individual families with their own Bonferroni screens, and underpowered tests are labelled as such, with direction-consistency reporting as the pre-declared remedy.

2.11 Statistical analysis

The primary segmentation endpoint is external query Dice. The primary classification endpoints are in-domain AUROC with the pre-declared grounding and error-propagation contrasts and, externally, rank discrimination (AUROC) plus the frozen specificity-0.90 operating point — the pooled operating point being the pre-declared cross-cohort test. Intervals bootstrap at the most independent available unit (image level in all current cohorts). Comparisons are paired: both models are scored on identical case-level resamples with two-sided sign-tail p-values, $p = 2\min(\Pr[\Delta \le 0], \Pr[\Delta \ge 0])$, and percentile 95% intervals;

operating-point contrasts resample the malignant cases under each model's frozen threshold; McNemar's χ^2 (continuity-corrected) accompanies the in-domain operating-point contrast. Resample counts: 5,000 in-domain, 2,000 external and pooled. Multiplicity is controlled per pre-declared family by Bonferroni (0.05 divided by family size): family T in-domain (six tests), families $T_c/T_d/E/F/P$ externally (five each), family G in the joint phase (four). Effect sizes accompany p-values throughout; significance is not interpreted as clinical usefulness without effect magnitude, calibration, error type, and external-domain behaviour.

2.12 Scope, reproducibility, and methodological boundaries

Explanation. The predicted mask is the primary localisation output; class-specific explanations target the reported classifier's decision score (the linear or joint head's logit, or the cosine margin of Eq. (8) for prototype heads). The five analysis axes are executed as the pre-registered Phase-6 battery (Section 2.7.3), and ground-truth masks serve explanation evaluation only unless an experiment is explicitly designated mask-supervised.

Deployment outlook. A deployment pathway was designed and then deliberately descoped. The frozen architecture admits a gradient-free mode in principle (supports encoded once, prototype vectors cached), and a distilled mobile student with FP32/FP16/INT8 exports would be benchmarked on size, memory, latency, and degradation relative to the teacher. No student model, quantised export, or latency measurement is reported; such results would be an offline edge-feasibility simulation, not a clinical device validation.

Reproducibility. Clean manifests are re-derived from raw files and hash-verified against the audit manifests, and every reported table is generated from saved predictions, not transcribed logs. The full artefact manifest — dataset versions, duplicate-drop lists, episode manifests, feature-bank caches, cached probabilities, label parsers, the TNUI polarity record, locked tolerance-reference tables, seeds, software versions, checkpoints — is listed in Supplementary Section S5, and Algorithm S1 (Supplementary Section S6) summarises the pipeline end to end. Target cohorts are never used for architecture, threshold, calibration, or explainer selection.

Boundaries. No ground-truth target mask enters ordinary end-to-end inference: predicted-mask descriptors are the primary condition; oracle regions appear only in the labelled oracle rows and the error-propagation analysis. Labels are not silently harmonised across provenance; the parsers, vote precedence, and TNUI polarity record are released artefacts. External cohorts are never used for fitting, tuning, thresholding, or calibration; pooling is limited to the single pre-declared pool under the replication gate; the raw-raw convention is fixed study-wide. The multi-source episodic extension, the uncertainty-aware prototype, the shrunk-covariance scorer, and the consistency loss are extension-level components; none is claimed in the primary pipeline.

3 Results

Ten pre-registered test families organise the inferential results. Table 3 maps each family's letter to its cohort, the research question it answers, and its phase; the prose uses short descriptive names, with letters retained in tables for precision. Per-cohort and operational tables are consolidated in the Supplementary Information (Tables S1–S22); the main text leads each phase with its consolidated or family-level table.

3.1 Cohorts, audits, and frozen operating conventions

Table 4 summarises the audited cohorts entering the classification battery; deduplication, cross-dataset hashing, and the official split files were applied as in Section 2. The TN5000 manifest re-derived byte-identically in every phase (3, 3b–3e), and the Phase-3d/3e replication gates reported zero drift on every

Table 3: Pre-registered test families: letter, cohort, motivating research question (Section 1.1), and phase. Test rosters and per-family Bonferroni screens are fixed in Section 2.11 and the phase protocols. The class-prototype label-budget sweep (Phase 3) and selective prediction (Phase 4) are pre-registered components reported outside the lettered families.

Family	Name used in prose	Cohort (unit)	RQ	Phase
T	in-domain TN5000 family	TN5000 test (image)	RQ3, RQ4, RQ6	3b
Tc	TNUI-test family	TNUI-2021 test (image)	RQ2, RQ3	3c
Td	TNUI-val family	TNUI-2021 val (image)	RQ2, RQ3	3d
P	TNUI-pool family	TNUI test+val pool (image)	RQ2, RQ3	3d
E	DDTI family	DDTI (image)	RQ2	3e
F	TNUI-train probe	TNUI-2021 train (image)	RQ2	3e
G	joint in-domain family	TN5000 test (image)	RQ5	4
H	joint pooled family	TNUI test+val pool (image)	RQ5	4
J	ThyUS2Path case family	ThyUS2Path (case)	RQ2, RQ3	5
K	explanations family	TN5000 + ThyUS-600 (image)	RQ8	6

replicated cell (POOLED_OK). Segmentation quality control preceded any classification table on every external cohort, against the Phase-2 zero-shot references (TNUI 0.7079, DDTI 0.6782 at K=20; per-cohort QC rows: Supplementary Table S1). Empty-mask fallback was rare in-domain (1.1% val / 1.2% test, Table 4).

Table 4: Classification cohorts after audit (locked). n = usable cases; prev. = malignant prevalence. TN5000 is the source domain; TNUI-2021 splits and DDTI are zero-shot external cohorts.

Cohort	Role	n	malignant (prev.)	Notes
TN5000 train	source fit	3448	2430 (0.705)	clinical split file train.txt
TN5000 val	thresholds/temperature	476	360 (0.756)	val.txt
TN5000 test	in-domain evaluation	950	693 (0.729)	test.txt
TNUI-2021 test	external (3c)	139	134 (0.964)	0 cross-dups vs TN5000
TNUI-2021 val	external (3d)	274	258 (0.942)	1 val–test cross-dup excluded
TNUI-2021 train	external probe (3e)	966	915 (0.947)	pooled policy n/a
DDTI (XML-labelled)	independent external (3e)	189	145 (0.767)	1 within-dup dropped

3.2 Supervised segmentation reference and external gap (Phase 1)

The full-data ResNet34-UNet reference (single seed, validation-only threshold 0.60, test-time augmentation, TTA) defines the annotation-expensive ceiling: the in-domain TTA Dice sits within the published TN3K band, and the external rows quantify the domain-shift cost before any few-shot machinery (full table: Supplementary Table S2). TTA effects are small and sign-mixed — the largest shift is +1.11 Dice points on TNUI-all, and DDTI moves slightly in the opposite direction — and DDTI is visibly the harder masked cohort (sensitivity 0.83, median HD95 roughly doubled), consistent with its lower zero-shot prototype scores below.

3.3 Few-shot support-conditioned segmentation (Phase 2)

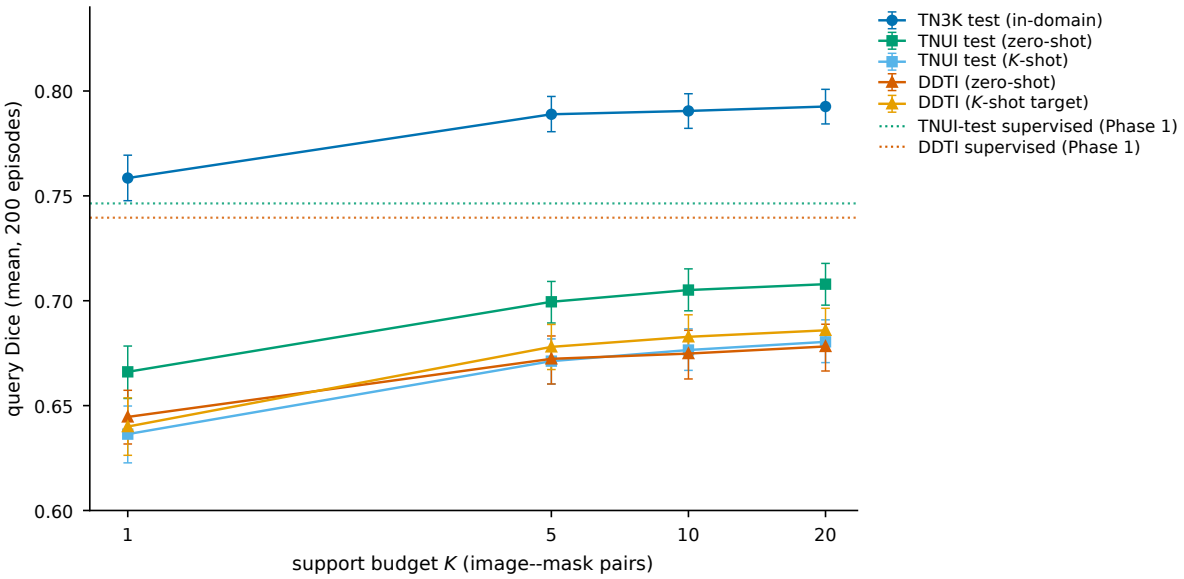


Figure 2: Few-shot segmentation saturates early and transfers without target annotation. Query Dice (fusion decoder) versus support budget K with nested episodes (200 episodes per cell, 1,600 query predictions; error bars: episode-blocked bootstrap 95% CIs). Dotted lines: the fully supervised Phase-1 external references. Zero-shot source-support transfer lands within 3.9 (TNUI-2021) and 6.1 (DDTI) Dice points of the supervised external reference at K=20 and already captures 98.6–99.5% of it at K=5; the parameter-free direct-matching variant (Supplementary Table S3) is 3.2–20.9 points lower in every cell.

Episodic evaluation (200 episodes per condition, Q_m=8, nested budgets) yields three findings (Table 5). (i) The learned fusion decoder outperforms parameter-free direct matching in all 20 condition–budget cells, by +3.2 to +20.9 Dice points (full rows: Supplementary Table S3). (ii) The annotation budget saturates early: K=5 already delivers 98.6–99.5% of the K=20 score in all conditions (Fig. 2). (iii) Zero-shot transfer with source supports lands within 3.9 Dice points of the supervised external reference on TNUI (0.7079 vs 0.7464) and within 6.1 on DDTI (0.6782 vs 0.7396), while using *zero* target annotations. Best validation episodic Dice 0.7776 (epoch 13; early stop 21; 37.7 min). The support-domain bias observed on TNUI (target supports underperform source supports) is reported as a limitation and motivates the joint-resourcing question of Phase 4.

Table 5: Phase-2 query Dice by support budget K_m (fusion decoder; locked). 200 episodes per cell (Q_m=8; 1,600 query predictions), episode-blocked bootstrap 95% CIs in parentheses. Last column: decoder advantage over parameter-free direct matching at K=20 (full direct rows in Supplementary Table S3).

Condition	K=1	K=5	K=10	K=20	Δdirect (K20, pts)
TN3K test (in-domain)	0.7585 (0.7477–0.7694)	0.7889 (0.7806–0.7974)	0.7905 (0.7822–0.7987)	0.7926 (0.7843–0.8008)	+11.7
TNUI test (zero-shot)	0.6661 (0.6534–0.6784)	0.6995 (0.6894–0.7092)	0.7051 (0.6952–0.7152)	0.7079 (0.6979–0.7178)	+20.9
TNUI test (K-shot)	0.6364 (0.6227–0.6498)	0.6712 (0.6603–0.6818)	0.6765 (0.6668–0.6866)	0.6804 (0.6705–0.6909)	+5.3
DDTI (zero-shot)	0.6446 (0.6317–0.6573)	0.6723 (0.6603–0.6832)	0.6748 (0.6627–0.6859)	0.6782 (0.6665–0.6888)	+11.0
DDTI (K-shot target)	0.6400 (0.6263–0.6537)	0.6780 (0.6672–0.6887)	0.6828 (0.6727–0.6933)	0.6859 (0.6760–0.6964)	+3.3

3.4 In-domain classification with frozen descriptors (Phases 3 and 3b)

All heads share the frozen Phase-1 encoder, the fixed K_m=20 TN3K support episode, and identical training regimens; thresholds (specificity $\geq$ 0.90 on raw validation probabilities) and temperatures are frozen from validation only. In-domain level rows (AUROC: ERM 0.9401, R1p 0.8902, oracle R1g 0.9105) and the per-model frozen operating points appear in Supplementary Tables S4 and S6; the pre-registered paired battery is Table 6. Lesion grounding beats the identical frozen whole-image head (T1: +0.0368, p=0.0004), and fine-tuning remains the in-domain rank ceiling (T2: +0.0498, p<0.0004). The secondary contrasts (relocated rows: Supplementary Table S5) price error propagation directly (T3: +0.0202, p=0.0048; sens@spec90 analogue T6: +0.0982, p<0.0004) and return two negative or null findings that we report as such: the context ring adds nothing (T4: +0.0074, p=0.1388; hypothesis H6 is null), and ERM dominates the in-domain operating point (T5: −0.0749, p<0.0004).

Two calibration readings are worth flagging in-domain: fitted temperatures are $T < 1$ for every frozen-descriptor head (mildly *under*-confident logits) but $T = 1.969$ for ERM (over-confident); and ERM is the best-calibrated model in-domain (Brier 0.0878, ECE 0.0634). Neither property survives the external shift, as the external battery shows below.

Error propagation is diffuse rather than confidence-localised (quartile table: Supplementary Table S7): the oracle-minus-predicted AUROC gap is modest in the lowest-confidence quartiles, vanishes in Q3, and is descriptively largest in the *highest*-confidence quartile (+0.0589), the mean gap (≈0.022) closely matching the global T3 contrast. Segmentation error therefore imposes a diffuse ≈2-point tax on grounded features, and there is no confidence-gated "safe subgroup" in which predicted masks are output-equivalent to oracle boxes.

The class-prototype label-budget sweep (K per class $\in \{5, 10, 20, 50, 200, \text{all}\}$, 25 draws; Supplementary Table S8) carries the classification-side annotation-efficiency claim: grounding dominates the identical whole-image budget at every K (+0.046 to +0.096 AUROC); R1p saturates by K=50 (0.6383 versus 0.6379 full-budget) and stays within 0.013–0.029 of the oracle, while R0 has not saturated even at K=200; and the ring is null-or-negative throughout (R2p $\leq$ R1p, R2g $\leq$ R1g at every K), corroborating the T4 null under a different head family. Absolute levels sit far below the linear heads (0.59–0.66 vs 0.85–0.91), as expected for a cosine prototype rule on frozen descriptors; the linear head remains the pre-declared primary classifier.

Table 6: Phase-3b paired tests (locked; 5000 paired bootstrap resamples, Bonferroni $\alpha = 0.0083$; McNemar χ^2 on discordant malignant cases). Δ = A−B. Empirical p-values are floored by the resample count: reports of p<0.0004 denote zero zero-crossings in 5000 resamples.

Test	Contrast	Metric	Δ (95% CI)	p	Sig.	Interpretation
T1	R1p − R0	AUROC	+0.0368 (0.0161, 0.0589)	0.0004	YES	grounding > image
T2	ERM − R1p	AUROC	+0.0498 (0.0308, 0.0691)	<0.0004	YES	fine-tune rank ceiling

Secondary contrasts T3–T6 (with the T5 McNemar pair), relocated row-for-row: Supplementary Table S5.

3.5 External classification battery (Phases 3c, 3d, 3e)

All external rows are computed with the frozen source objects; thresholds and temperatures come from TN5000 validation only, and no target case informed any decision. Per-cohort families are Bonferroni-screened at $\alpha = 0.01$; the single pre-declared pool is TNUI test+val with its own family. Table 7 consolidates the matrix: all cohorts, all six models.

TNUI-2021 test (Phase 3c).

n=139 (134 malignant). Grounding degrades least: the AUROC shift drop shrinks monotonically with lesion-centring strength, from ERM's 0.423 down to R2g's 0.052. The fine-tuned ERM operating point does not transfer (T5c: −0.157, p=0.0028) — an inversion of the in-domain T5 (+0.075 ERM advantage) — and ERM sits significantly below the true-mask oracle (T4c: −0.3170, p=0.0004); T1c–T3c are power-limited (5 benign cases). Rows: Supplementary Table S9.

Second cohort and pooled test (Phase 3d).

TNUI-2021 val (n=274, 16 benign) discriminates poorly for *every* model including true-mask oracles (0.577–0.624), revising the friendly-cohort reading of 3c; the replication gate passed with zero drift. The pre-declared TNUI-pool family found the grounded heads' operating-point advantage over ERM replicated and significant (P5: −0.120, CI −0.181.. − 0.061, p<0.0004); the rank-transfer flip of 3c did not replicate (P2 null). Rows: Supplementary Table S10.

Independent-institution counterexample and heterogeneity probe (Phase 3e).

DDTI (n=189, 44 benign) sits at chance for *all* models — including below-0.5 true-mask oracles (0.414–0.422) — and the operating point inverts: ERM wins (E5: +0.1456, p=0.0012). TNUI-train (n=966) behaves like val (≈0.55–0.65), confirming 3c's test split as the friendly cohort; burned-in DDTI acquisition annotations are a documented confound (qualitative grid). Rows: Supplementary Table S11.

External verdict (as locked).

(i) In-domain ranks are fixed (Table 7, first row). (ii) External rank discrimination is cohort-dominated: identical frozen rules swing the true-mask oracle over 0.42–0.85 (Table 7). (iii) The grounded operating point (Supplementary Figure S1) transfers across the TNUI institutional family (test p=0.0028; train and pooled p<0.0004; val in direction) and *reverses* on the independent DDTI cohort (p=0.0012): an institutional-family transfer claim with a documented counterexample. (iv) Calibration does not survive the prevalence shift in any model (external ECE 0.25–0.42); rank metrics therefore remain primary externally, and deployment claims require local recalibration.

Table 7: Cross-cohort evidence matrix (AUROC; locked rows). Sign consistency of R1p−ERM: AUROC +, −, +, +; sens@spec90 +, +, +, − across TNUI test/val/train and DDTI. Oracle (R1g) AUROC externally: 0.834/0.613/0.646/0.422.

Cohort	ERM	R0	R1p	R2p	R1g (oracle)	R2g (oracle)
TN5000 test (in-domain)	0.9401	0.8534	0.8902	0.8829	0.9105	0.9029
TNUI test	0.5172	0.5791	0.7119	0.6776	0.8343	0.8507
TNUI val	0.6237	0.5901	0.5775	0.6008	0.6126	0.5770
TNUI train	0.5936	0.5472	0.6316	0.6441	0.6459	0.6078
DDTI	0.4746	0.5116	0.4967	0.5017	0.4216	0.4143

3.6 Joint partial-supervision fine-tuning (Phase 4)

Phase 4 asked whether the segmentation pathway can be kept intact while a classification head is fine-tuned into the shared trunk, and whether doing so closes the in-domain fine-tuning gap (T2). Protocol v1 (single-stage joint fine-tune at 10^{-4}, AUROC-only selection) failed the pre-registered segmentation non-collapse gate outright (TN3K-test Dice 0.6686 against a gate of 0.7966; triage showed the fresh head's

gradients rotating the shared encoder from the first epoch). Protocol v2 — trunk-frozen head warmup, a low-rate (10^{-5}) joint phase, and seg-guarded checkpoint selection on the validation split only — passed both non-collapse gates (JOINT-g 0.8258, JOINT-img 0.8165 against the same 0.7966 gate), with frozen-row integrity exact (ERM 0.9401, R1p 0.8902).

The joint in-domain family G (Table 8; level rows: Supplementary Table S12) answers the motivating question in the negative, and we report it as such: joint training did *not* close T2 — it widened it (G1: −0.0703, CI −0.0927.. − 0.0486, p<0.0004). JOINT-g also sits *below* the frozen grounded head it was meant to surpass (G2: −0.0205, CI −0.0365.. − 0.0045; p=0.0124, clearing the family bar by 10^{-4} — significant but fragile). The one large positive is mechanistic: under identical multi-task conditions the grounded head beats the whole-image head by 14.9 points (G3: +0.1489, CI 0.1136..0.1848) — the grounding advantage survives joint optimisation; it is not an artefact of freezing. At the pre-frozen operating point the joint model loses outright (G4: −0.2376 CI −0.2785.. − 0.1962; McNemar 41 vs 206 on test positives, χ^2=108.89; G2–G4 rows: Supplementary Table S5). JOINT-g had not saturated at the pre-registered 15-epoch cap (validation AUROC was still improving; the cap was not extended), so family-G magnitudes are conservative.

The pair of runs documents a precise dilemma: closing the T2 gap requires the task-specific representational reorganisation ERM performs from scratch, and any such reorganisation is precisely what a segmentation non-collapse constraint forbids. v1 unknowingly chose one horn (fine-tuning performance without an intact segmenter); v2 proves the other (an intact segmenter without the fine-tuning premium).

The external family H completes the picture (Table 8; level rows: Supplementary Table S13). The primary external test is nominal only: on the single locked TNUI test+val pool JOINT-g exceeds ERM's sensitivity at the frozen operating point by 6.6 points (H1: +0.0661, 95% CI 0.005..0.128, p=0.033, beyond the family bar; McNemar 86 vs 60). The rank test is a flat null (H3: −0.011, p=0.858; secondary rows: Supplementary Table S5), and H2 is significant: joint grounding transfers its operating point *worse* than the frozen R1p by 5.3 points of pooled sensitivity (H2: −0.0534, CI −0.097.. − 0.010, p=0.011). The pooled operating-point ordering, R1p ≈ 0.666 > JOINT-g ≈ 0.612 > ERM ≈ 0.546, is the distilled Phase-4 verdict. The joint segmenter travels adequately (external Dice 0.674–0.765, empty-mask ≤ 0.3%), so the deficit lives in classification head-room, not mask collapse. Every Phase-4 strand converges on the same operational choice: *the frozen-descriptor R1p pipeline is the primary LG-PM model.*

The pre-registered selective-prediction component (Supplementary Table S14; Supplementary Figure S2) shows calibrated-margin abstention biting at the frozen operating point without any refitting: retaining the most-confident 70% of TN5000-test cases lifts JOINT-g's sensitivity at spec90 from 0.564 to 0.840, and at 50% coverage the curve reaches 0.874 in-domain and 0.897 on the external TNUI pool. The effect is referral-shaped, not accuracy-shaped: approximately 86–87% of the frozen point's malignant misses sit inside the abstained lower-confidence half on both panels (261 of 302 misses in-domain; 132 of 152 on pooled TNUI). No threshold or temperature was re-estimated for these numbers (raw-raw; the validation-fitted T=0.523 supplies only the confidence margin).

3.7 Case-level external validation on ThyUS2Path (Phase 5)

Phase 5 carried the frozen pipeline onto ThyUS2Path, a histopathology-labelled cohort from an acquisition network disjoint from all source pools, under the audit-gated licence protocol (Section 2.7.2) — the study's central external-validation demand: a second, pathology-verified institution, evaluated at case level. The audit passed end to end: label coverage was complete for Batch 1 (6,005/6,005) and near-complete for Batch 2 (2,495/2,503), giving an overall parse rate of 0.9991 (8,500/8,508); the surviving join matched the corpus's own image map exactly (5,893/5,893 Batch-1 stems). Assembly gave 842 cases (601 + 241), and

Table 8: Phase-4 pre-registered families. G (in-domain TN5000 test; 5,000 paired bootstrap resamples, Bonferroni α=0.0125): *locked*. H (pooled TNUI test+val, n=413, 392 positives, 2,000 resamples, α=0.0167): *locked*. Δ = A−B; empirical p floored as p<0.0004 (zero zero-crossings in 5,000 resamples). H1's 95% CI excludes zero while p=0.033 exceeds the family bar (the ≈98.3% CI includes zero): reported as nominal.

Test	Contrast	Metric	Δ (95% CI)	p	Sig.	Question
G1	JOINT-g − ERM	AUROC	−0.0703 (−0.0927, −0.0486)	<0.0004	YES	is the T2 fine-tune gap closed? — **No: widened**
H1	JOINT-g − ERM	sens@spec90 (pooled)	+0.0661 (0.0051, 0.1276)	0.033	no	PRIMARY external op. point — nominal only

McNemar (H1, pooled positives n=392): JOINT-g-right/ERM-wrong 86 vs ERM-right/JOINT-g-wrong 60, $\chi^2 = 4.28$ (nominal $p \approx 0.039$) — direction consistent, family bar not cleared.

Secondary contrasts G2–G4 (with the G4 McNemar pair) and H2–H3, relocated row-for-row: Supplementary Table S5.

conservative whole-case near-duplicate screening excluded three, leaving 839 cases and 8,465 images (598 + 241), of which 522 are malignant (prevalence 0.622, inside the $[0.20, 0.85]$ gate). Frozen-row integrity held exactly (ERM re-hit 0.9401; the retrained R1p head reproduced 0.8889 against the locked 0.8902, within the 0.002 integrity tolerance; both JOINT arms matched to $\leq 10^{-6}$, temperatures 0.523/0.200), and segmentation QC was healthy (empty masks on 10.9% of images, 926/8,465; JOINT-g decoder 3.5% (293/8,465)). Mean predicted nodule areas were 4.1% and 5.9% of the frame respectively. The cohort ships without lesion annotation, so no external Dice is computable and the oracle regimes are excluded as pre-declared. Label polarity was operator-attested on the seeded audit grid, whereupon the inference licence was granted; the licence upgrade unsealed the significance column and altered no number.

Ranking collapses universally (per-model rows: Supplementary Table S15; shift matrix: Supplementary Table S16): all six models land between AUROC 0.478 and 0.548 on the case unit, only ERM's interval clearing 0.5 — by eight thousandths. AUPRC spans 0.608–0.666 against the 0.622 prevalence, and shift drops from source are 0.31–0.39 AUROC for every linear head and 0.39 for ERM (drop 0.3921, the largest); even JOINT-img, modest in-domain (0.7207), loses 0.24, and the first-list-image view shows the same pattern (0.481–0.533; Supplementary Table S17). Two observations place the collapse at cohort level rather than in the LG-PM machinery: it afflicts ERM — a pure whole-image classifier with no segmentation in the loop — at least as severely as any descriptor head, and it persists when case aggregation is bypassed entirely. Calibration does not travel either: source-fitted temperatures *worsen* external ECE for five of six models (post-calibration ECE 0.26–0.31; only ERM improves, 0.2813 → 0.2145), and Brier scores (0.29–0.35) sit far above the source regime.

The pre-registered case family J replicates the operating-point story at full strength (Table 9). At the frozen specificity-≥0.90 thresholds, the grounded descriptor retains 15.9% case-level sensitivity where the fine-tuned ERM retains 1.3% (J1: +0.1457, 95% CI 0.1149..0.1782, p<0.0004; McNemar 77 vs 1, $\chi^2 = 72.12$, $p < 10^{-4}$). In integers: of the 83 malignant cases R1p catches at the frozen point, ERM misses 77; of the 7 ERM catches, R1p misses 1 — an eleven-fold relative advantage, and the external case-level replication of the internal operating-point result. Frozen-over-joint transfer replicates with larger magnitude (J3: −0.0978, CI −0.1264.. − 0.0709, p<0.0004), and the case-level operating-point ordering, R1p (0.1590) > JOINT-g (0.0613) > ERM (0.0134), mirrors the pooled Phase-4 ordering (0.666 > 0.612 > 0.546) across an entirely disjoint cohort. The rank contrasts are null at the family bar (J2 p=0.0436; J4 p=0.1104; rows: Supplementary Table S5). The honest reading is twofold: absolute performance on ThyUS2Path is far from diagnostic grade for *every* model — 15.9% sensitivity at 90% specificity is a comparative, not a clinical, statement — and the cohort functions as a stress test of what survives a severe acquisition and case-mix shift; what survives is precisely the lesion grounding.

Table 9: Phase-5 pre-registered family J (ThyUS2Path case level, n=839, 522 malignant; 5,000 paired case-bootstrap resamples; Bonferroni α=0.0125): *locked*. Δ = A−B; empirical p floored as p<0.0004 (zero zero-crossings in 5,000 resamples). Computed under the sealed audit-gate discipline and unsealed only after A1–A4 passed and the operator attested label polarity (A5); the licence upgrade altered no number in this table.

Test	Contrast	Metric	Δ (95% CI)	p	Sig.	Question
J1	R1p − ERM	sens@spec90	+0.1457 (0.1149, 0.1782)	<0.0004	YES	PRIMARY external op. point — **replicated**
McNemar (J1, malignant cases n=522): R1p-right/ERM-wrong 77 vs ERM-right/R1p-wrong 1, $\chi^2 = 72.12$ ($p < 10^{-4}$) — 77:1 discordants in favour of the grounded head at the frozen point.						

Secondary contrasts J2–J4, relocated row-for-row: Supplementary Table S5.

3.8 Lesion-grounded explainability battery (Phase 6)

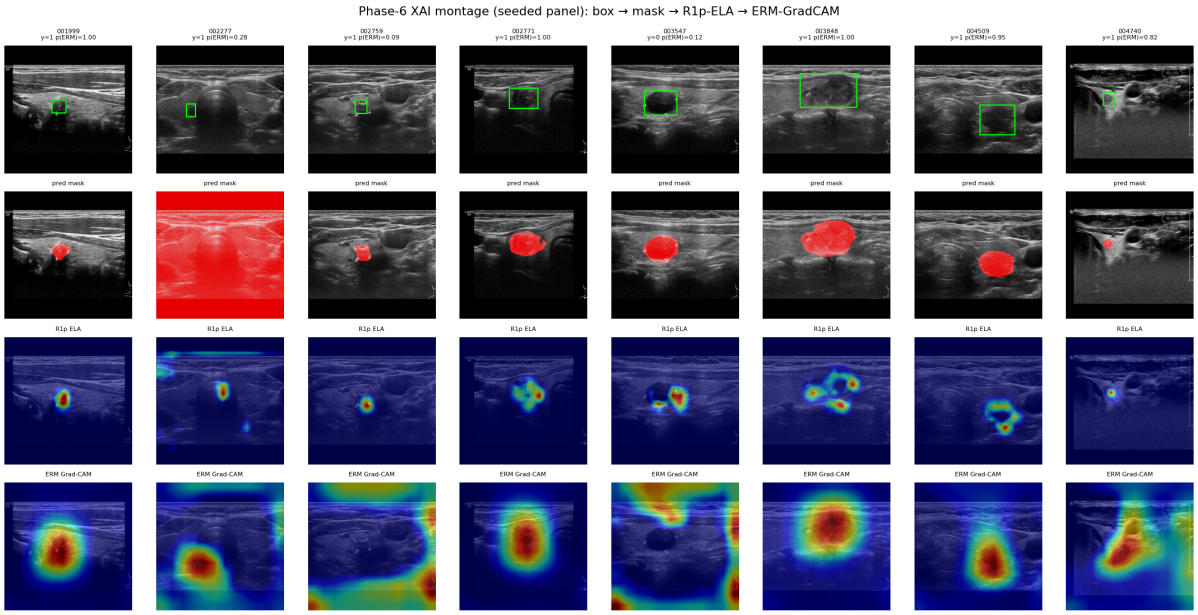


Figure 3: Lesion-grounded explanations on representative cases. Rows: TN5000-test and ThyUS-600 cases (labels given). Columns: raw B-mode frame with expert box; frozen predicted mask; R1p-ELA; Grad-CAM (with the model's probability). The grounded attribution is confined to the nodule by construction; Grad-CAM energy disperses into gland tissue and acquisition artefacts, visibly more so on the external rows.

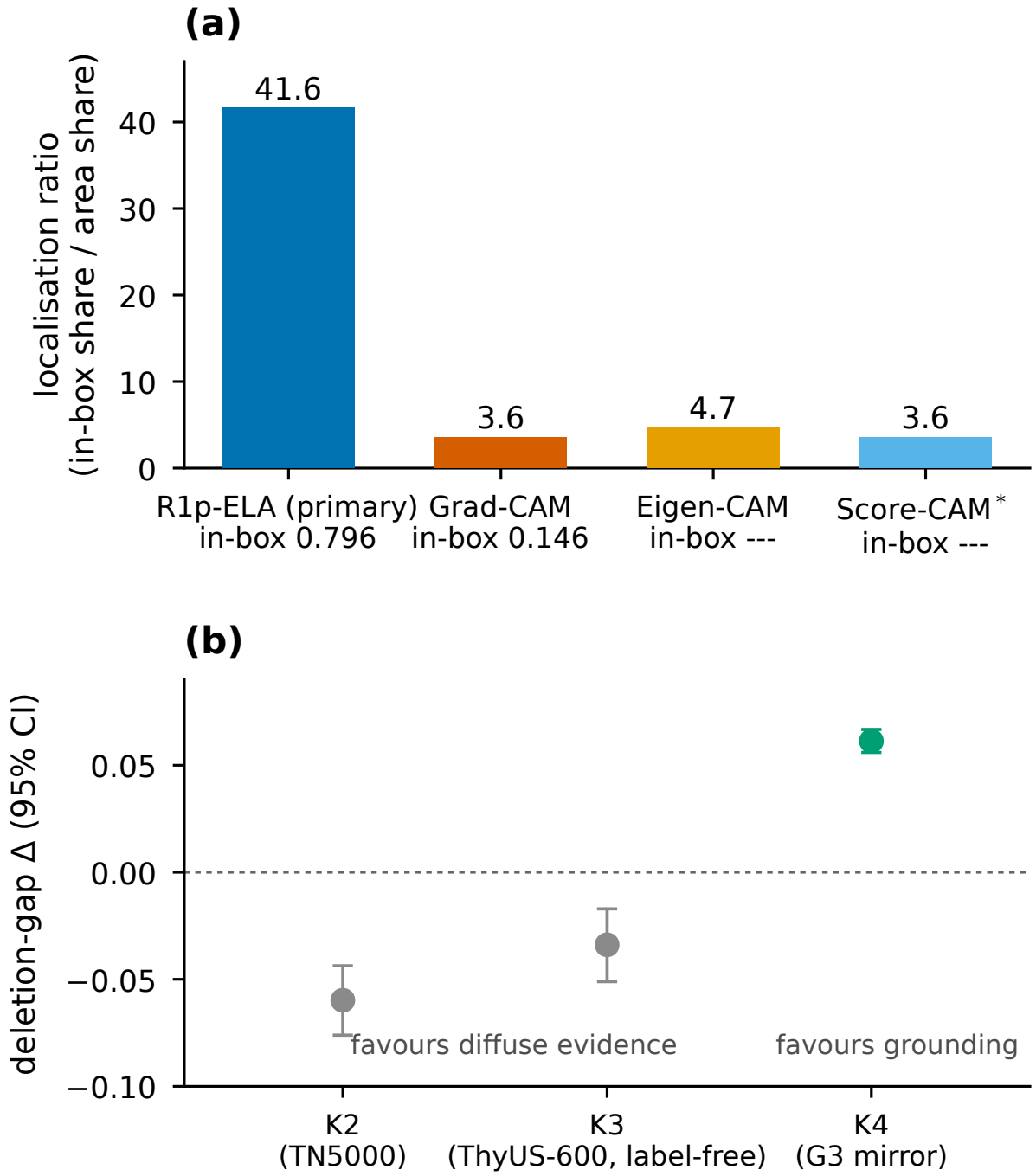


Figure 4: Explanation quality by axis (Phase 6, family K). (a) Localisation against TN5000 expert boxes: R1p-ELA concentrates an order of magnitude more of its mass on the nodule than Grad-CAM, Eigen-CAM, or Score-CAM (paired Δ K1 = +38.0, CI 35.35..40.83, p<0.0004); in-box share printed per bar (*Score-CAM on the seeded n=150 subsample). (b) Deletion-gap deltas show the complementary, metric-driven face: K2/K3 are significantly negative (removal-based faithfulness rewards diffuse evidence; mechanism in text), while within the joint pair the grounded head wins as the mechanism predicts (K4). CIs: 5,000-resample paired bootstrap; Bonferroni α=0.0125.

The battery ran unmodified on TN5000-test (n=950, 693 malignant, expert boxes on all 950 images) and the seeded ThyUS-600 first-image panel (n=600, 376 malignant; drawn from the licensed 839-case cohort and triangulated: 522 malignant, 8,465 images, prevalence 0.6222). Integrity structure held end-to-end: live

probabilities reproduce the Phase-3/4 archives (ERM 0.9404 vs 0.9401; R1p 0.8889 vs 0.8889; JOINT arms to $\leq 2.25\times10^{-5}$ max $|\Delta p|$ against the 10^{-4} gate), and the 32-image ELA exactness probe confirmed the closed-form logit decomposition (max deviation R0 1.6×10^{-6}, R1p 2.9×10^{-7}, R2p 2.7×10^{-7}, JOINT-g 7.8×10^{-8}, JOINT-img 2.6×10^{-8}; gate 10^{-3}). All inferential statements use 5,000 paired image-level resamples at Bonferroni α=0.0125 (seed 42); empirical p=0.0 is typeset p<0.0004. A seeded qualitative panel accompanies the quantitative tests (Fig. 3).

The primary test is decisive (Table 10; Fig. 4; per-map statistics: Supplementary Table S18). Against expert-annotated nodule boxes, R1p-ELA attains a localisation ratio of 41.6 (median 27.8) where Grad-CAM attains 3.6: a paired separation of K1 $\Delta = +38.00$ (95% CI 35.35..40.83, p<0.0004). The gap is not an artefact of the comparator: Eigen-CAM and Score-CAM sit at 4.7 and 3.6, an order of magnitude below the grounded explanation. The decomposition marks R1p's precision as segmenter-grade (79.6% of attribution mass inside the box, normalised entropy 0.717) against ERM-family diffusion (in-box mass 0.146, entropy 0.928). The pointing game cuts the other way descriptively (Eigen-CAM 0.736 > R1p 0.637 > Score-CAM 0.580 > Grad-CAM 0.514): where a diffuse map happens to peak inside the box it scores like a concentrated one; the ratio and in-box-mass statistics instead report where the *bulk* of the explanation lies. Under shift, ERM's attention deserts the nodule: Grad-CAM's energy inside the frozen predicted mask falls from 0.1253 in-domain (≈3.2× the mask's area share) to 0.0549 on ThyUS-600 (≈1.3×).

The faithfulness pair is honestly null-to-reversed for the intended direction, and we report it as such (K2: −0.0598, CI −0.0761.. − 0.0437, p<0.0004 on TN5000; K3: −0.0339, CI −0.0511.. − 0.0171, p<0.0004 as the label-free external replication; rows: Supplementary Table S22). Inspection shows the reversal is carried by the random-order control, not by the guided curves: explanation-guided deletion degrades both models essentially identically (areas 0.3334 vs 0.3315), while random deletion erodes the diffuse image-wide classifier more slowly (random-control areas 0.4986 vs 0.5564; deletion/insertion areas: Supplementary Table S19). The deletion-gap statistic rewards redundant, spatially spread evidence rather than explanation quality per se — the known sensitivity of removal-based metrics to the model's evidence distribution. Insertion (descriptive, n=300) favours ERM (0.2043 vs 0.0351) with the same proviso, compounded by the out-of-distribution nature of sharp-lesion-on-blur hybrids for a mask-conditioned head. Within the joint pair, however, grounding helps exactly as the G3 mechanism predicts (K4: +0.0613, CI 0.0560..0.0667; Supplementary Table S22), the deletion-gap ladder rising monotonically with the degree of built-in grounding (JOINT-img 0.041 < JOINT-g 0.102 < R1p 0.165).

Stability is mixed and instructive (Supplementary Table S20): under intensity noise R1p-ELA is the more stable explanation in both domains (mean Spearman 0.921 vs 0.832 on TN5000; 0.799 vs 0.755 on ThyUS-600), while affine warps invert the ordering (0.771 vs 0.871; 0.640 vs 0.810) — the two rows are the same concentration geometry read in opposite directions; concentrated maps are penalised by per-pixel correlation after geometric displacement, additive noise taxing diffuse maps more. These axes were pre-declared non-gated and are reported descriptively.

The randomisation cascade produced the battery's one failed gate, and we report it as such (Supplementary Table S21). R1p, whose design admits only a head-level stage (the frozen trunk is the shared measurement instrument), retains mean $|\rho|$= 0.2991 — the gate fails against the pre-registered $|\rho|$<0.2 bar. ERM decorrelates below the bar only at its deepest stage (0.1205), and at the matching shallow stage (fc only, 0.2948) it fails identically: a 0.2 threshold at readout depth is unattainable for either model, rather than evidence of explanation defect. Two follow-up diagnostics, whose decision rules were committed before they were run, localise the result. Independent resampled heads correlate at only 0.0920 on the same images, so the retained 0.2991 sits +0.2072 above that floor: the pixel map is *not* head-agnostic geometry.

Meanwhile, the per-feature logit attribution decorrelates completely under head resampling (Spearman −0.0150, |−| 0.0191; Pearson −0.0016). Combined with the exactness gate, the operational picture is precise: the ELA attribution is the exact, weight-specific decomposition of the linear head in feature space, and its pixel-space rendering is partly bound by the shared in-lesion feature field — which is precisely the lesion grounding the architecture exists to provide, and which K1 shows is where the explanation correctly lives.

Table 10: Phase-6 pre-registered family K (image-level paired bootstrap, 5,000 resamples, seed 42; Bonferroni α=0.0125; two-sided empirical p, with p=0.0 typeset p<0.0004; point estimates are bootstrap-distribution means — for the heavy-tailed K1 ratio the mean of replicate means (38.0047) deviates from the arithmetic mean of differences (38.0270) by 0.0223, ≈1.1× the Monte-Carlo standard error, as expected). K3 is label-free and therefore independent of the Phase-5 licence. Significance column: Bonferroni verdict.

Test	Metric	A	B	Δ(A−B) (95% CI)	Sig.	Note
K1	localisation ratio	R1p-ELA	ERM-GradCAM	+38.0047 (35.3461..40.8339)	YES	PRIMARY plausibility

Secondary contrasts K2–K4, relocated row-for-row: Supplementary Table S22.

4 Discussion

This study asked whether few-shot, support-conditioned nodule segmentation can improve cross-domain thyroid-ultrasound malignancy classification while providing calibrated, anatomically grounded, and faithful explanations (Section 1.1). The answer across six executed and locked phases is deliberately three-part. *Anatomical grounding: yes, decisively* — the grounded explanation is an order of magnitude more lesion-aligned than any unconstrained-CAM comparator, and the grounded operating point replicates on an independent pathology-confirmed case-level cohort with an eleven-fold sensitivity advantage over the fine-tuned classifier. *Calibration: no* — no model's calibration survives prevalence shift, and we report this rather than recalibrating on target data. *Faithful: yes in feature space, reversed on removal-based metrics* — and the reversal itself carries a methodological lesson about what deletion-style statistics measure.

Few-shot segmentation and annotation efficiency (RQ1, RQ2; H1, H2).

Few-shot support conditioning delivers near-ceiling nodule segmentation with almost no annotation, in-domain and zero-shot. A learned fusion decoder beats parameter-free prototype matching in all 20 condition–budget cells (+3.2 to +20.9 Dice points at K=20); the parameter-free variant collapses to 0.47–0.57 Dice under zero-shot transfer, so the contribution sits in the decoder, not in nearest-prototype matching of the PANet kind [21–23]. The budget saturates early: K=5 nested support masks already deliver 98.6–99.5% of the K=20 score in every condition (Table 5), a direct quantification of dense-label efficiency (Gap 1); and zero-shot transfer with *source* supports lands within 3.9 (TNUI-2021) and 6.1 (DDTI) Dice points of the fully supervised external reference at zero target annotation (Supplementary Table S2) [2,3]. H1 (annotation efficiency) is therefore *supported*, and in a stronger form than posed: the support mechanism substitutes for target annotation entirely. H2 is *not supported in its strong form*: the few-shot model narrows but does not eliminate the supervised external gap. Two honest qualifications apply. The support-domain bias on TNUI — target supports underperform source supports (0.6804 vs 0.7079 at K=20) — inverts the intuition that target-side support always helps; we report it as a limitation rather than tune it away. DDTI is the hardest masked cohort for both paradigms (supervised 0.7396; zero-shot 0.6782), consistent with its doubled boundary error (median HD95 ≈22 vs 10–11 px) and burned-in annotations.

Does grounding improve classification? (RQ3, RQ6; H3, H6).

Grounding improves the frozen operating point substantially, cohort-wide ranking not at all, and calibration never. In-domain on TN5000-test [5], the grounded linear head beats the identical whole-image head (T1: +0.0368 AUROC, p=0.0004), and the grounded label-budget curve dominates the image budget at every support size, saturating by 50 labels per class and staying within 0.013–0.029 of the ground-truth-box oracle (Supplementary Table S8). The fine-tuned ERM remains the in-domain rank and operating-point ceiling (T2, T5) — a fact we retain rather than minimise, because the external phases invert it: At the frozen specificity-0.90 operating point the grounded head's advantage replicates across the TNUI institutional family (pooled P5: +0.120 sensitivity over ERM, p<0.0004; third cohort F5: +0.126, p<0.0004), reverses on the independent DDTI cohort (E5: −0.146, p=0.0012; burned-in annotations documented), and replicates again — at case level, against a pathology gold standard, on 839 ThyUS2Path [6] cases untouched by any tuning — with J1 +0.1457 (p<0.0004; McNemar 77 vs 1). *Rank* discrimination does not travel: external AUROCs are cohort-dominated (even true-mask oracles span 0.41–0.85), and on ThyUS2Path every model collapses to 0.478–0.548 with AUPRC at the 0.622 prevalence floor. H3 is thus *supported for the operating point, null for ranking, and refuted for calibration*: the grounding signal concentrates at the frozen operating point — the quantity fixed before external data were seen — while ranking is dominated by site-specific case-mix structure no frozen rule escaped (Table 7; cf. retrospective multi-cohort thyroid CAD [35]). RQ6 receives a clean null: the peri-lesional context ring adds nothing (T4: +0.0074, p=0.139), replicating across the prototype sweep at every budget under a different head family. H6 is *refuted*, twice — constraining the ROI+context premise for nodule classification [36]: on frozen descriptors, context contributes neither discrimination nor label efficiency; it merely dilutes what is already lesion-local.

Error propagation (RQ4; H4).

Replacing expert boxes with predicted masks costs a small, bounded, and diffuse tax, and no confidence gate can waive it. T3 prices the replacement at ≈2 AUROC points (+0.0202, p=0.0048) and T6 at ≈10 operating-point points. Crucially the tax is *diffuse*, not catastrophic: stratified by predicted-mask confidence, the oracle-minus-predicted gap does not localise to bad segmentations and is descriptively largest in the *highest*-confidence quartile (Supplementary Table S7). H4's first clause (a small, budget-shrinkable gap) is supported; its second (uncertainty identifies the worst cases) is *not*. For a pipeline claim this is the useful form: predicted-mask classification deploys end-to-end at a known, bounded cost, and mask confidence should not be used to waive caution.

Partial supervision and the joint dilemma (RQ5; H5).

An intact segmenter and the fine-tuning premium cannot be had at once: the dilemma, not either protocol, is the finding. The naive joint fine-tune failed its own pre-registered non-collapse gate (Dice 0.669 against a gate of 0.797); the corrected protocol (frozen-head warmup, 10^{-5} joint phase, seg-guarded selection) preserved the segmenter and passed both gates. The intact segmenter is *then* what sacrifices the fine-tuning premium: the T2 gap widened (G1 −0.0703), the joint model sat below its frozen counterpart (G2 −0.0205), lost outright at the frozen operating point (G4; McNemar 41 vs 206), and transferred externally worse than the frozen grounded head (H2 −0.0534). Yet under identical joint conditions, grounding beat multi-tasking by +0.1489 AUROC (G3; Table 8): the grounding, not the freezing, carries the signal. H5 is *refuted in this instantiation*, and the refutation is informative: closing the fine-tuning gap requires exactly the representational reorganisation that a segmentation non-collapse constraint forbids.

Uncertainty, calibration, and referral (RQ7; H7).

Calibration never survives the shift, but confidence-based referral still removes most of the frozen operating point's malignant misses. External ECE sits at 0.25–0.42 for every head after prevalence shift; on ThyUS2Path five of six models deteriorate *further* after source-fitted temperature scaling, so deployment-level use requires local recalibration. What does work, with nothing refitted, is selective prediction on the calibrated margin: retaining the most-confident 70% lifts in-domain sensitivity at spec90 from 0.564 to 0.840, with the external pool tracking closely (Supplementary Table S14); at 50% coverage, external sensitivity reaches 0.897. The mechanism is referral-shaped — 86–87% of the malignant cases missed at the frozen point sit inside the abstained lower-confidence half on both panels — so abstention selectively clears threshold-proximal malignancies rather than globally inflating accuracy. H7 is *supported*: a small referral stream (≈30% of cases) removes the majority of the frozen point's false negatives without touching a single parameter.

Explanation reliability (RQ8; H8).

The pre-registered battery (family K) gives H8 a split verdict by clause, and we regard the split as a methodological contribution. *Lesion overlap: decisive support.* Against expert boxes, the grounded, exact linear attribution attains a localisation ratio of 41.6 against 3.6 for Grad-CAM [28] (K1 +38.0, CI 35.35..40.83, p<0.0004), with Eigen-CAM and Score-CAM controls [37,38] at 4.7 and 3.6 — the separation is a property of grounding, not of Grad-CAM's weakness. The descriptives cohere: 79.6% of the grounded map's mass lies inside the expert box versus 14.6% for Grad-CAM, whose agreement with the predicted nodule falls from ≈3.2× area share in-domain to ≈1.3× on ThyUS-600: under shift, the fine-tuned classifier's attention deserts the lesion; the grounded mechanism's definitionally cannot. *Perturbation stability: partial support* — the grounded map is the more stable under intensity noise in both domains, the less stable under affine warps (Supplementary Table S20), the expected signature of concentrated versus diffuse maps. *Deletion/insertion faithfulness: not supported as posed.* K2 and K3 are significant in reverse (−0.0598, −0.0339), with identical explanations under guidance but divergent random-ordering controls: removal-based faithfulness here rewards redundant, spatially spread evidence rather than explanation fidelity, the documented sensitivity of this metric family to evidence distribution [39–41]. Inside the joint pair, where the confound cancels, grounding wins exactly as the mechanism predicts (K4 +0.0613). The randomisation cascade adds the final honest note: the readout-level R1p cascade fails its gate (0.2991 against the 0.2 bar — which the ERM readout-only stage, 0.2948, fails equally); committed diagnostics show the retention is neither head-agnostic geometry (random-vs-random floor 0.092; retained elevation +0.2072 above it) nor a broken explanation (per-feature attribution decorrelates completely, $\rho = -0.015$; exactness $\leq 1.6\times10^{-6}$). H8 is therefore *supported on overlap, partially on stability, and refuted as stated on removal-based faithfulness* — the hypothesis's conjunction, rather than the pipeline, is what the measurements reject. The take-away is to pre-register plausibility and faithfulness separately and interpret each against its explanation's geometry [29]; a pooled "trustworthy CAM" claim, of the kind thyroid CAD papers routinely display only qualitatively [1,42–44], would obscure exactly that split.

Cross-cutting interpretation.

Three themes unify the six phases. *First, the operating point is the unit of transfer*: every robust external result lives at thresholds frozen before external data were touched (P5, F5, J1); every cohort-wide rank comparison yields cohort-dominated heterogeneity instead — post-hoc threshold selection would have written a different, irreproducible paper. *Second, the external-validity gradient is informative, not*

embarrassing: transfer holds within the TNUI family, inverts on DDTI under a documented annotation confound, and re-emerges on ThyUS2Path at case level against the strongest label standard; a single pooled external AUROC would have concealed all of that. *Third, pre-registered gates are content, not ceremony*: the failed gates — the joint protocol's segmentation collapse (v1), the cascade's R1p stage, the DDTI counterexample, H6's double null — shaped the model choice (frozen R1p as primary), the XAI interpretation, and the scope of the claims. This is the reproducibility practice Gap 8 calls for, applied to our own pipeline first.

Limitations and future work.

Several limitations bound the claims. (i) External *absolute* performance on ThyUS2Path is far from diagnostic grade for every model (15.9% sensitivity at 90% specificity for the best head); the cohort is a stress test of what survives shift, and all external comparisons are relative statements at frozen operating points. (ii) The zero-shot empty-mask fallback rate on ThyUS2Path runs at 10.9% of images (22% of case-first images), so a nontrivial external minority is scored by the whole-image fallback pathway. (iii) TNUI-2021 benign counts are small (5 test, 16 validation), limiting cohort-level precision, and DDTI carries burned-in acquisition annotations as a documented confound. (iv) K1 evaluates against boxes rather than pixel masks, so the localisation reward is at box granularity; and the qualitative evidence is a montage, not a reader study — human-usefulness evaluation with clinicians remains open. (v) The frozen-descriptor heads are confidence-under-scaled in-domain ($T<1$) and all models require local recalibration before any deployment interpretation. (vi) Phase 1's reference is single-seed, and the joint model had not saturated at its pre-registered epoch cap, making family-G magnitudes conservative. (vii) The cascade gate failure means pixel-space attribution maps should be read as partly bound to the in-lesion feature field even though their feature-space decomposition is exact and weight-specific; we have bounded, not eliminated, that ambiguity. The natural next experiments follow directly: the descoped edge-feasibility pathway (RQ9/H9) admits a gradient-free deployment mode with cached prototype vectors, to be benchmarked as an offline simulation rather than a clinical device; mask-uncertainty estimators that *do* localise classification risk (the present confidence quartiles do not); and a prospective multi-institution evaluation with per-site calibration, which the audit-gated protocol is designed to absorb without modification.

In conclusion, we present a lesion-grounded, prototype-based few-shot framework (LG-PM) for annotation-efficient thyroid-ultrasound diagnosis, evaluated across a fully pre-registered, audit-trailed programme. The central question receives a disciplined answer. Support-conditioned segmentation transfers nodule localisation to unseen ultrasound domains with zero target annotation, coming within a few Dice points of the fully supervised external reference and saturating after a handful of support masks. The predicted masks feed a classification pathway whose advantage is real and specific: lesion grounding beats identical whole-image heads in-domain, and — uniquely in this study — its operating-point advantage replicates externally, culminating in an eleven-fold case-level sensitivity advantage over fine-tuned classification on the pathology-confirmed ThyUS2Path cohort. That advantage does not extend to cohort-wide ranking, calibration, or removal-based faithfulness, each of which we report as measured: rank transfer collapses under shift for all models including oracles, calibration requires local refit, deletion-style metrics reward diffuse-evidence models, and one randomisation gate failed and is reported as failed. The framework does not diagnose cancer, and its external ranking and calibration limits are the honest boundary of the claim. What it establishes is that in thyroid ultrasound, lesion grounding is a transferable operating-point

advantage with anatomically aligned, exact, and audit-trailed explanations — delivered at a fraction of the annotation cost of supervised adaptation.

Supplementary Materials: The supplementary material is available online, comprising Tables S1–S22 and Figures S1–S2 referenced throughout the Results, plus extended methodological detail, a reproducibility artefact manifest, and an algorithmic summary (Algorithm S1).

Conflicts of Interest: The authors declare no competing interests.

Funding Statement: The authors received no specific funding for this work.

Availability of Data and Materials: All datasets are publicly available from their original publications (cited in the text). No new imaging data were generated.

Code Availability: Notebooks, seeds and audit artefacts for every phase are archived in the project ledger. Code is available from the corresponding author upon reasonable request.

Ethics Approval: All cohorts analysed are public, de-identified datasets (TN3K/TN5000, TNUI-2021, DDTI, ThyUS2Path); no new human data were collected. Ethics approval was not required for this secondary analysis.

Acknowledgement: Not applicable.

Author Contributions: All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed collaboratively. All authors read and approved the final manuscript.

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